Necessitation preserves validity

The Open Logic Project contributors

openlogic:normal-modal-logic/syntax-and-semantics:local-snapshot-20261013; CC BY 4.0

Editorial scope. Statement and complete author proof are included verbatim as the span lines around ollabel prop:Nec-rule of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- Kripke models and the box truth clause at a world
- validity as truth at all worlds of all models

Proposition 0.1. *If $!A$ is valid, then so is $\Box!A$.*

Proof. Assume $\models !A$. To show $\models \Box!A$ let $M \models= \langle W, R, V\rangle$ be a model and $w \in W$. If Rww' then $M \models !A[]w']$, since $!A$ is valid, and so also $M \models \Box!A[]w]$. Since $M \models$ and w were arbitrary, $\models \Box!A$. □

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